



FlowGrid

FlowGrid: Adaptive Traffic Light Control System

Real-Time Intersection Optimization Empowered by
Computer Vision & Deep Reinforcement Learning

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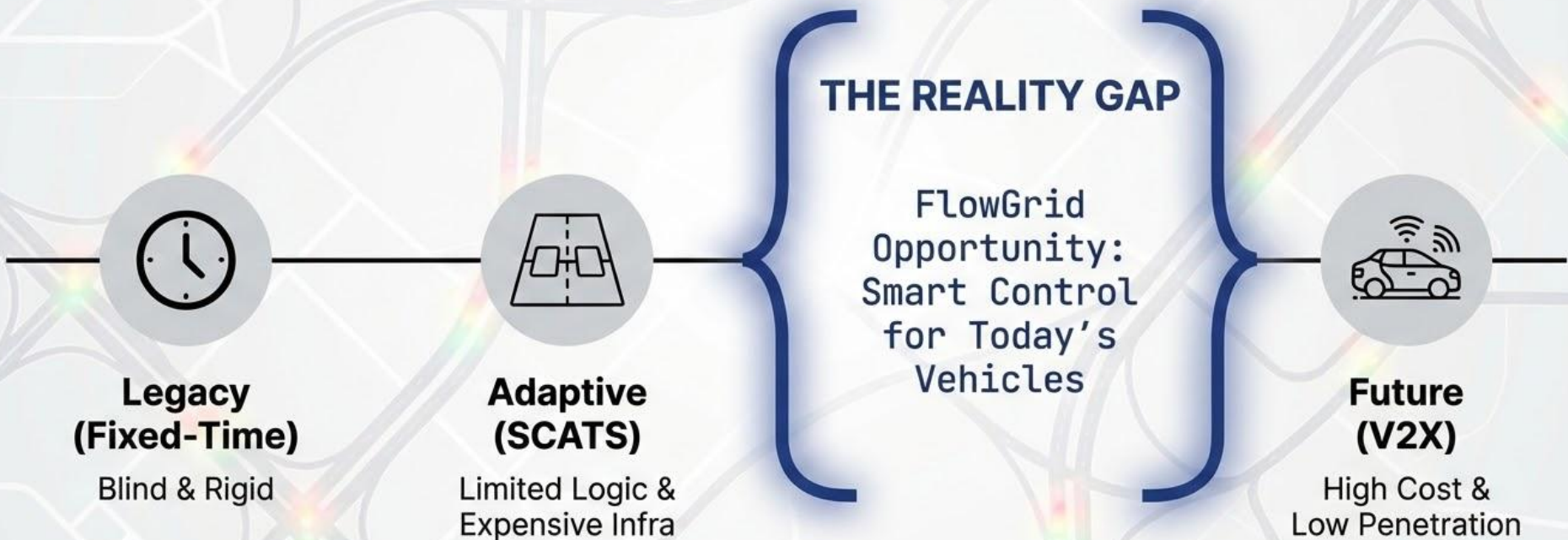
Group code: 26-1-D-30 (Phase A)

The Reality: Static Infrastructure

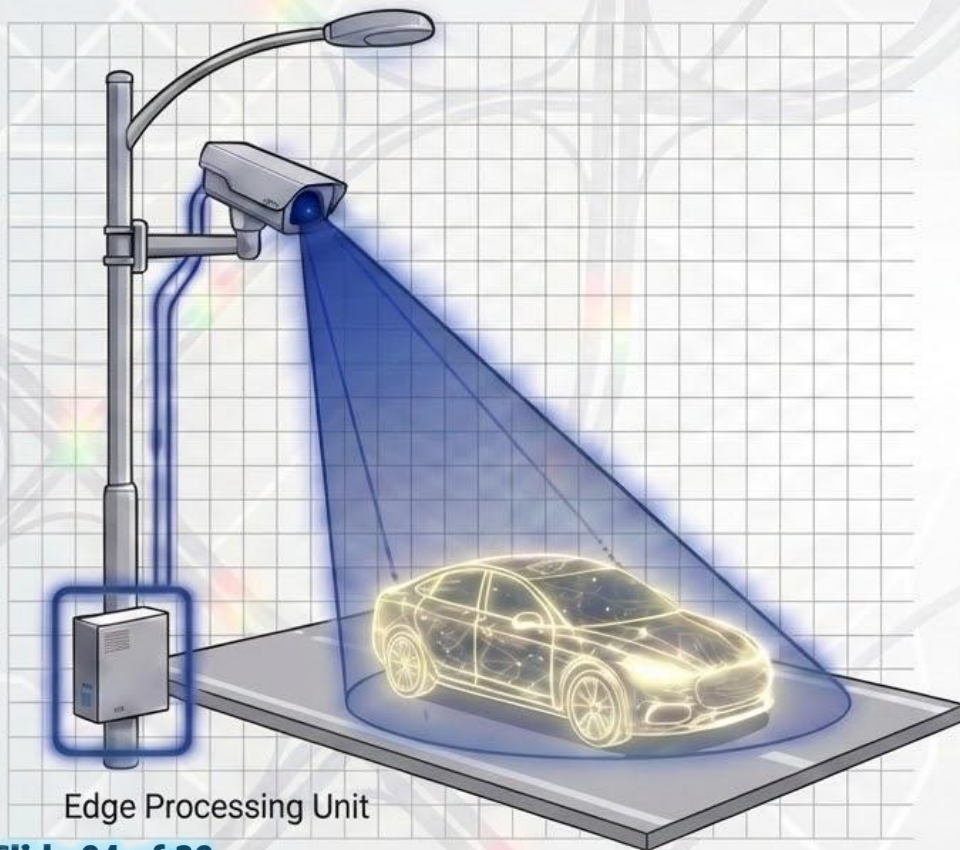
- ⚠ **Economic Impact:** Billions lost in fuel and productivity annually.
- ⚠ **Environmental Toll:** High CO2 emissions from inefficient stop-and-go cycles.
- ⚠ **Critical Failure:** Delayed emergency response times when seconds count.
- ⚠ **Root Cause:** Systems are 'blind' to real-time parameters.



Current Landscape & The Gap



The Solution: Camera-as-a-Sensor



Perception: Computer Vision (YOLO) replaces ground sensors

Leveraging real-time object detection models to identify and count vehicles without embedded road loops.



Tracking: DeepSORT (Identity Retention)

Maintains unique Vehicle IDs across frames to track trajectories and prevent double-counting, ensuring data integrity.



Decision: Deep Reinforcement Learning (DQN) optimizes flow

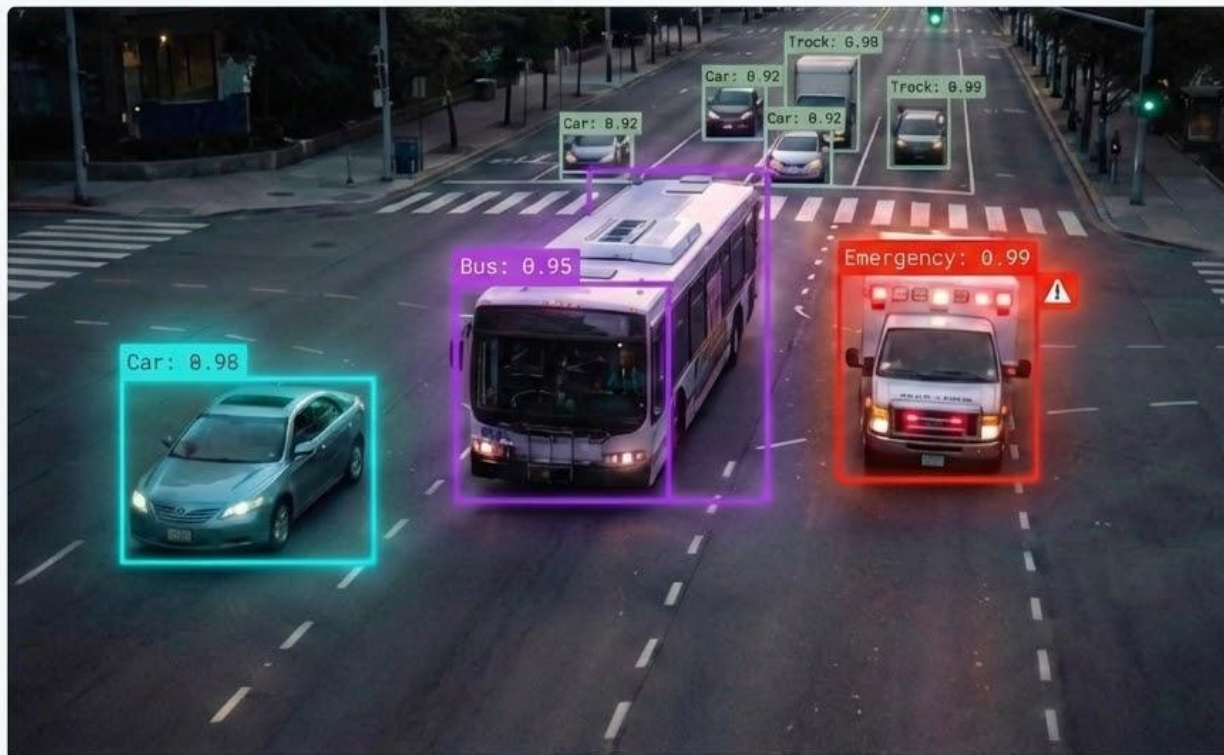
AI agent learns optimal signal timing strategies based on live traffic density and wait times to minimize congestion.

Methodology: Scalable Generalization Strategy

- **The Challenge:** Scalability. We cannot realistically model every city intersection individually.
- **The Strategy:** Randomized **Training**. We expose the RL agent to highly diverse, randomly generated environments in SUMO.
- **The Goal: Universal Logic.** The agent learns fundamental traffic dynamics, enabling zero-shot adaptation to new junctions.



Perception Layer: YOLOv8



Architecture: YOLOv8
(You Only Look Once)



Inference Speed:
45+ FPS



Latency:
< 100ms



Classification:
Private / Transit
/ Critical



Precision:
High

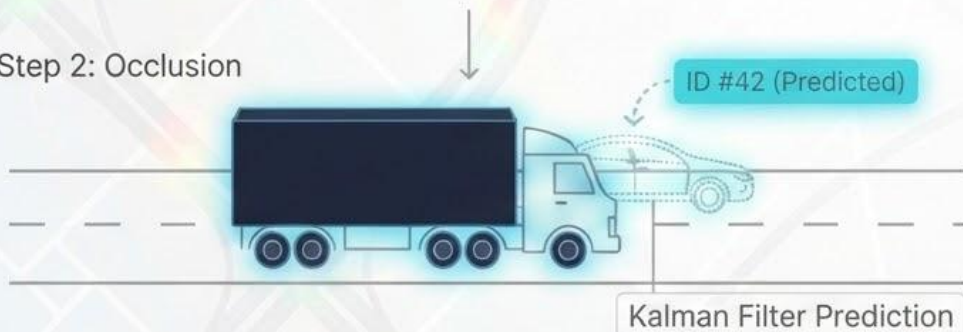
Tracking: DeepSORT Logic

Occlusion Handling

Step 1: Object Visible



Step 2: Occlusion



⚠ The Problem:

Detection sensors have no memory.

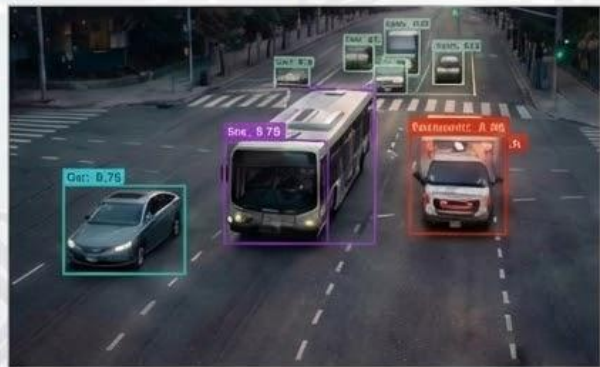
⊗ The Solution:

Kalman Filters predict object position during occlusion.

🕒 Result:

Continuous "True Waiting Time" calculation for every vehicle.

The Brain: Deep Reinforcement Learning (DQN)



State Space



Reward Function = Maximize Throughput - (Cumulative Waiting Time)

Key Functional Requirements



Autonomous Decision Making

System independently determines signal phases based on real-time density (RL Policy).



True Waiting-Time Optimization

Calculates individual vehicle wait-times to prevent starvation and ensure fairness.



Emergency Preemption

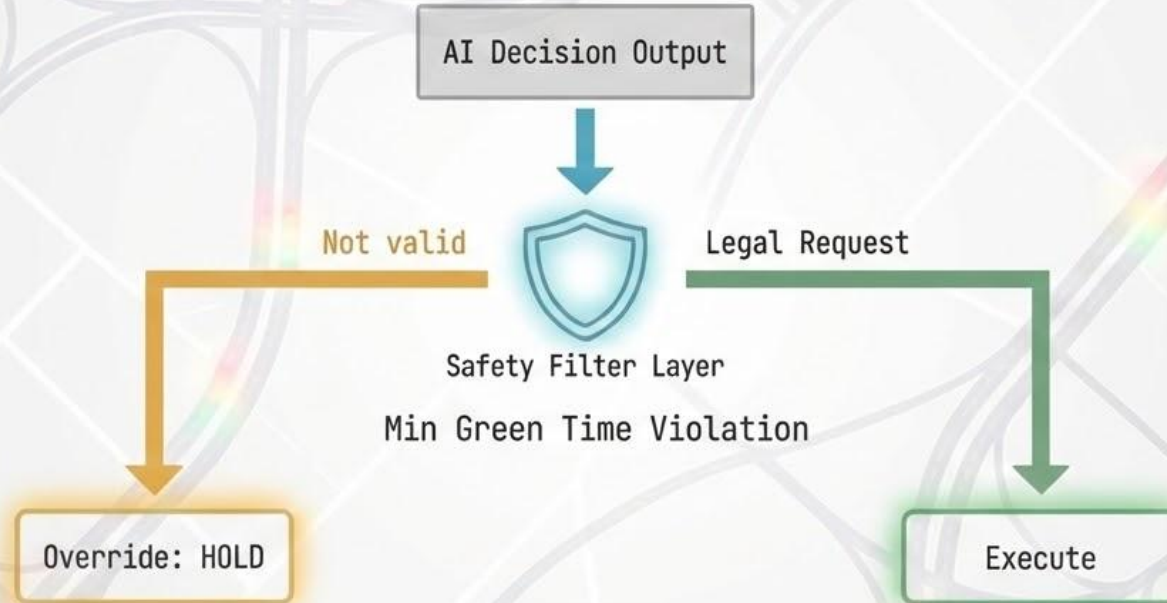
Immediate detection of **emergency vehicles** triggering a "**Green Wave**" priority protocol.



Fail-Safe & Safety Validation

Automatic fallback to **Historical Traffic Profiles** upon sensor **failure**, ensuring flow continuity based on learned patterns.

Inter Display: Safety & Reliability Architecture



Deterministic rules always supersede Probabilistic AI.

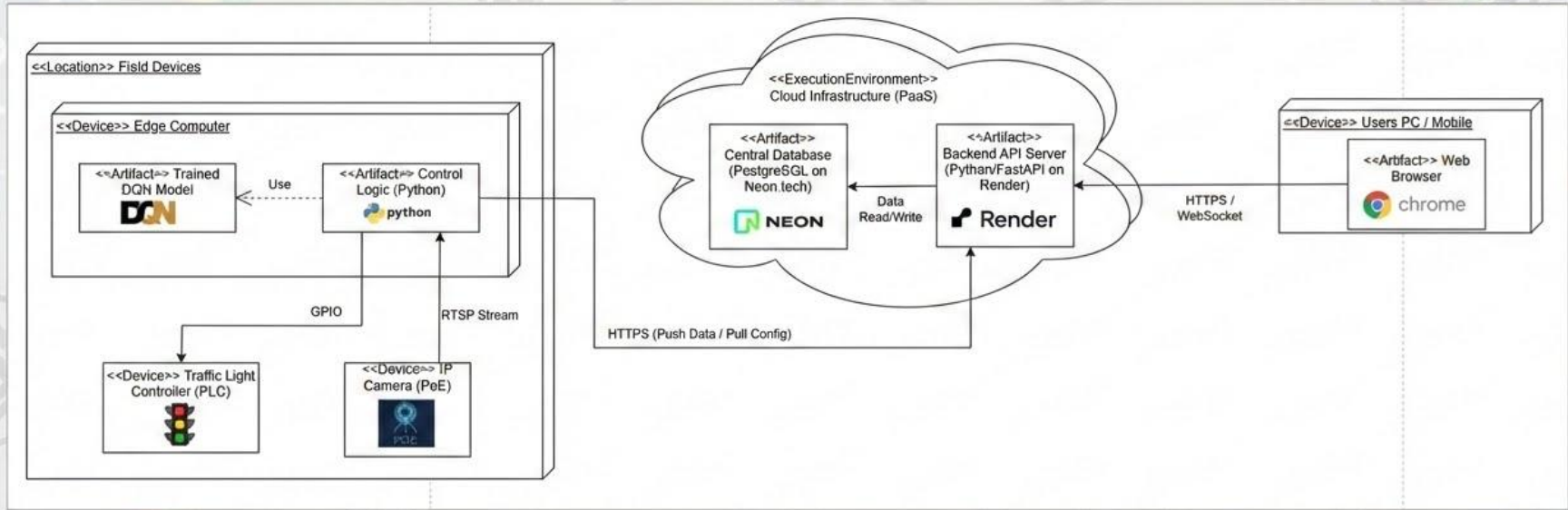
Emergency Priority Protocol



Result:
0.0s Delay

Priority Vehicle Detected
Traffic Signals Adjusted
Clear Passage Granted

Deployment Architecture



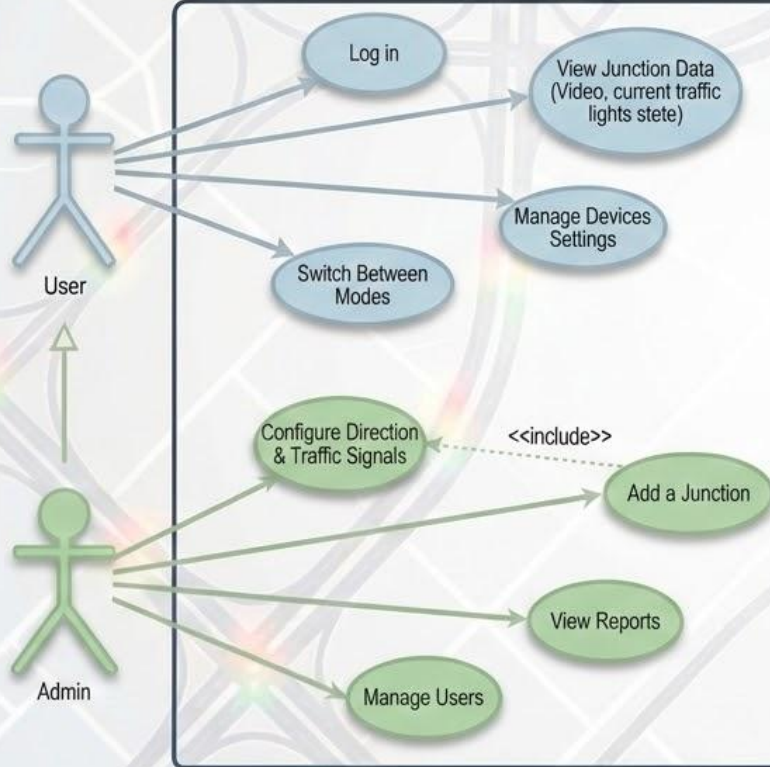
 **Edge Unit:** Industrial PC runs DQN logic with zero latency.

 **Cloud Layer:** Scalable Backend (FastAPI) & Database (PostgreSQL).

 **Protocols:** RTSP (Video), GPIO (PLC Control), WebSocket (Live UI).

 **Hybrid Model:** Decouples real-time control (Edge) from storage (Cloud).

Functional Design: User Roles



Two Actors: Standard User (Operator) vs. Admin (Manager).



User Privileges: Live monitoring & Mode switching (AI/Manual)

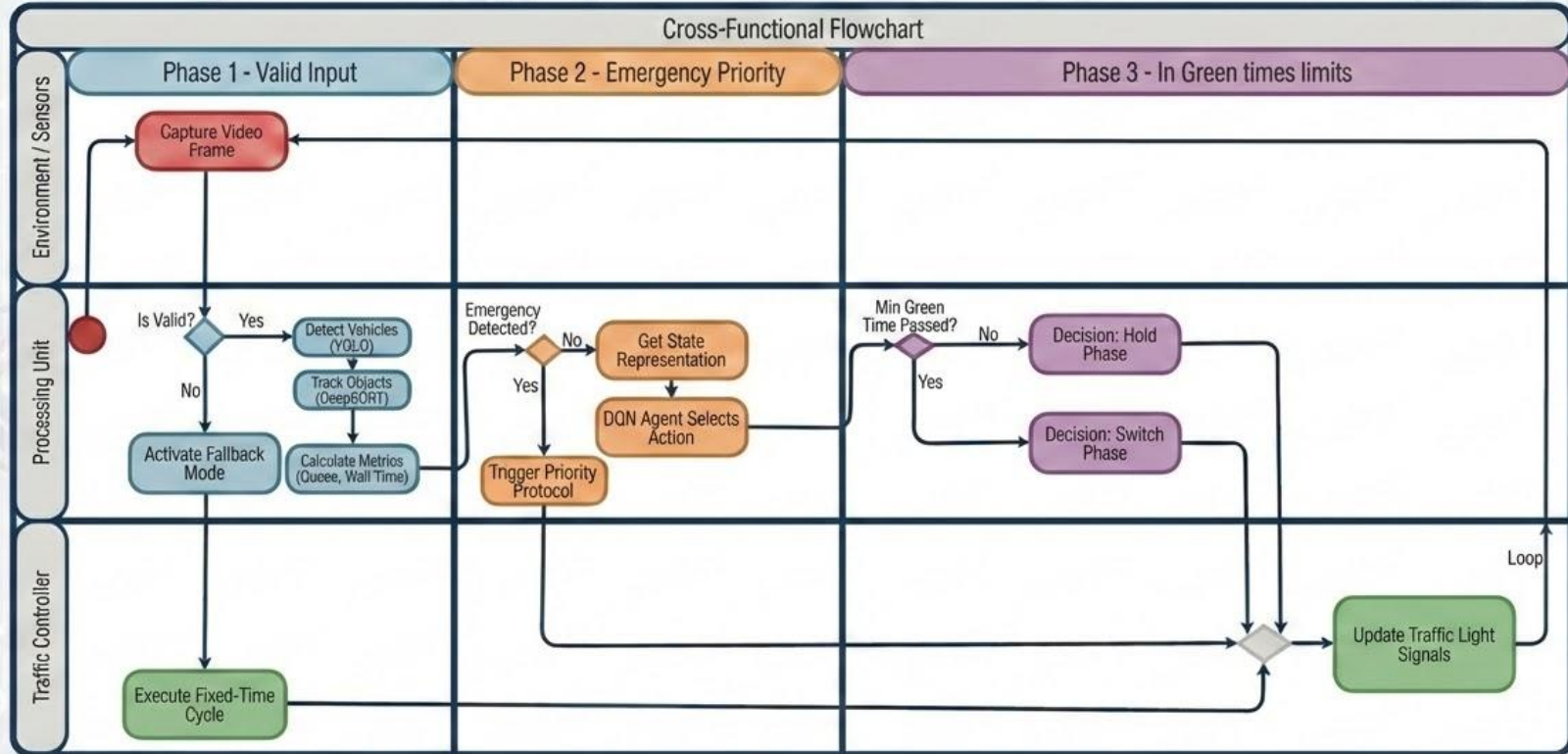


Admin Exclusives: User management, Reports, & System Config.



Dependencies: "Add Junction" forces Signal Configuration (<<include>>)

System Logic Flow



Operator Dashboard

Smart Traffic Spills Central

77746642ScafeldCagnebedvz77/theme-light

Smart Traffic System - Fully Synchronized

Convertible Slide

NORTH SOUTH

EAST WEST

Light	Wait	Cars
● RED (Stop)	25s	3 cars

Light	Wait	Cars
● RED (Stop)	25s	3 cars

Light	Wait	Cars
● GREEN (Go)	0s	19 cars

Light	Wait	Cars
● GREEN (Go)	0s	13 cars

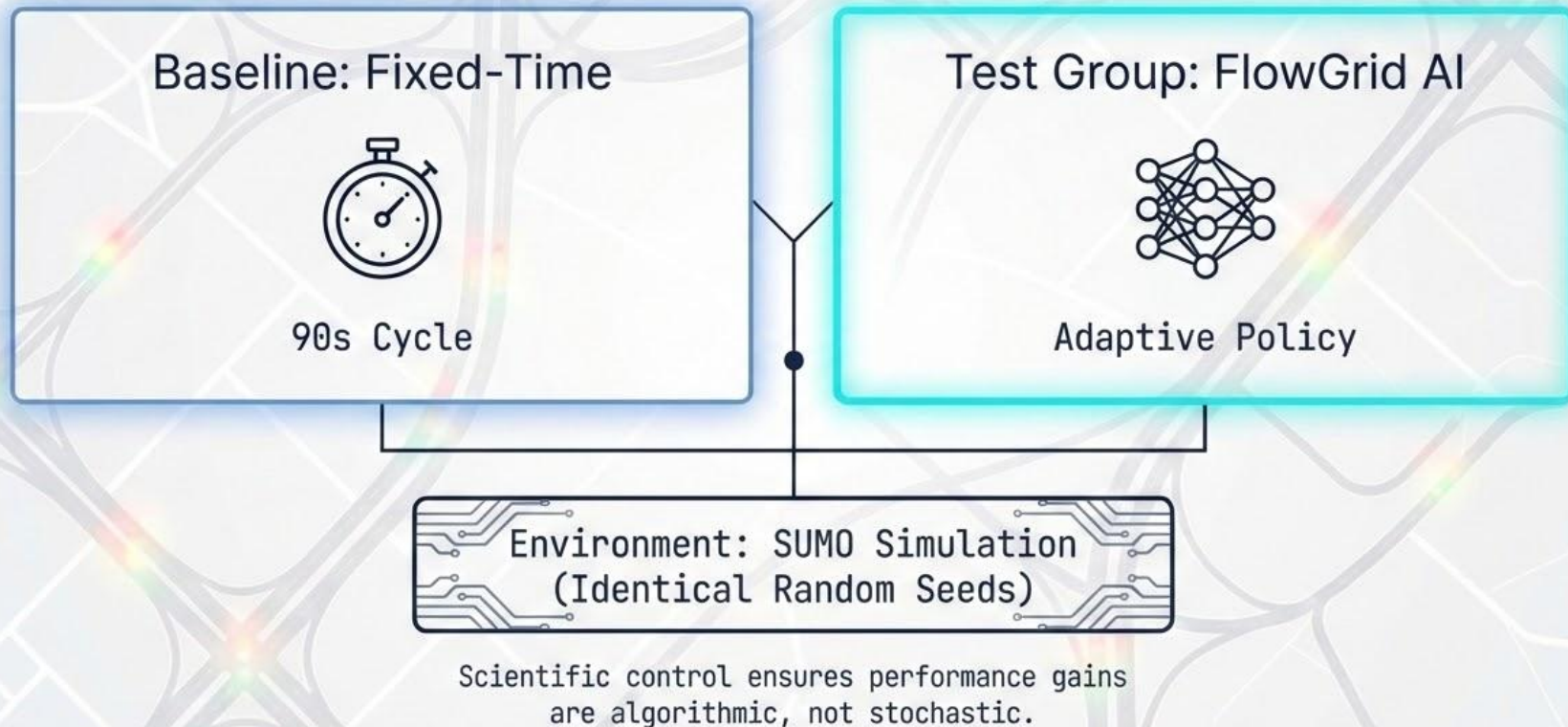
Unisila API - Built with GridDer - Settings

Real-time
RTSP Stream

WebSocket
Connectivity &
Manual Override

Live Queue &
Wait Telemetry

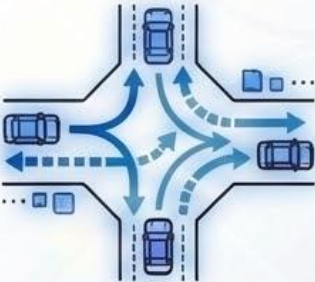
Verification Strategy: A/B Testing



ENGINEERING SIMULATION SCENARIOS: High-Fidelity Model Test Cases

STATUS ACTIVE

VERSION 1.3



**SCENARIO 1:
URBAN TRAFFIC
CONGESTION**

Simulating high-density vehicular flow, bottlenecks, and rush hour conditions.

STATUS ACTIVE

VERSION 1.3



**SCENARIO 2:
EMERGENCY RESPONSE
OPTIMIZATION**

Evaluating routing efficiency for priority vehicles under time constraints.

STATUS ACTIVE

VERSION 1.3



**SCENARIO 3:
SENSOR FAILURE &
DATA LOSS**

Testing system resilience and error handling in degraded perception inputs.

STATUS ACTIVE

VERSION 1.3

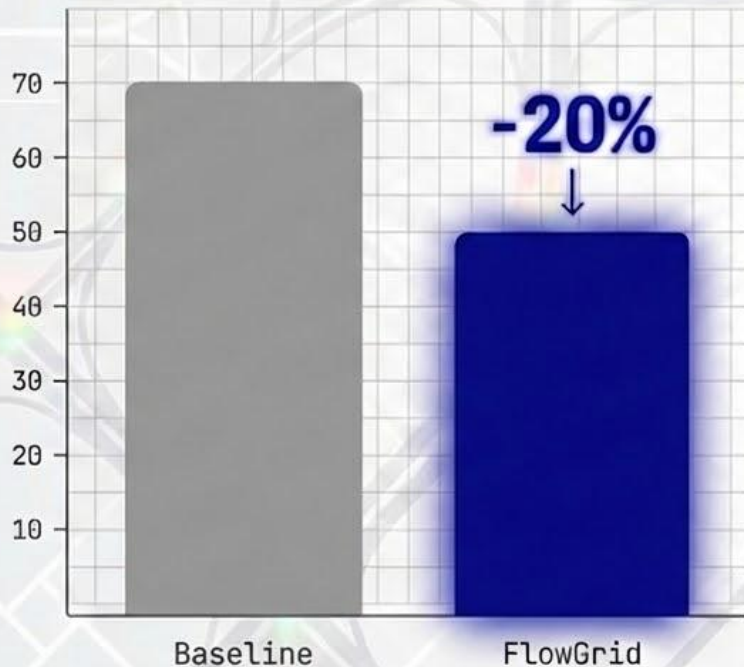


**SCENARIO 4:
OCCLUSION & VISIBILITY
OBSTRUCTION**

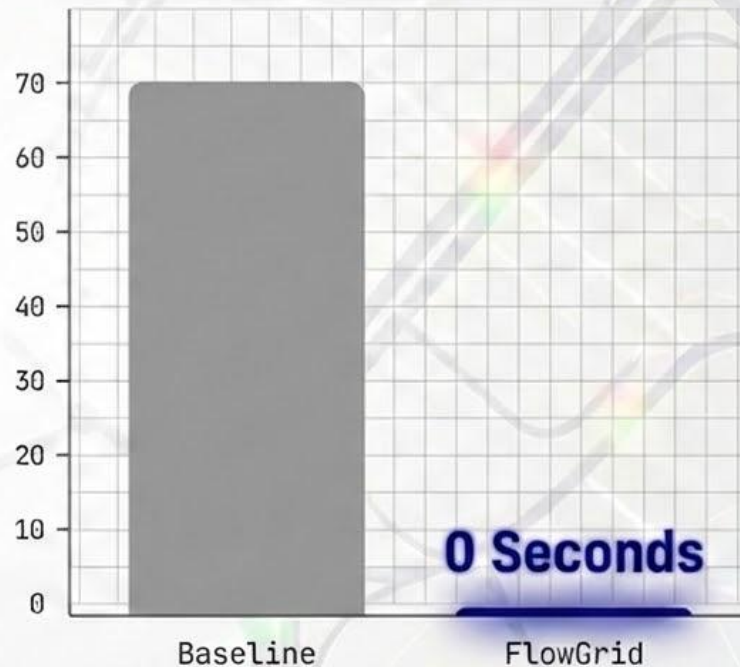
Assessing detection algorithms under severe line-of-sight challenges by large objects.

Expected Results

Avg. Wait Time (Heavy Traffic)



Emergency Vehicle Delay



ANTICIPATED ENGINEERING CHALLENGES:

Sim-to-Real, Hardware, & RL Stability



SCENARIO 1: SIM-TO-REAL GAP

- Domain Gap in Perception Models
- Dynamics Mismatch (Tire/Surface)
- Sensor Noise & Calibration



SCENARIO 2: HARDWARE CONSTRAINTS

- Compute Constraints (Edge Devices)
- Thermal Management
- Reliability & Durability



SCENARIO 3: RL STABILITY ISSUES

- Convergence Issues
- Safety during Training
- Generalization to Unseen Scenarios

Conclusion & Future Impact



- **Phase A Status:** System Architecture & Design fully completed.
- **Impact:** A cost-effective, AI retrofit for existing infrastructure.
- **Next Step:** Code Implementation, DQN Training, and System Integration.